

Synapses

Modelling presynaptic and postsynaptic mechanisms

Computational Neuroscience

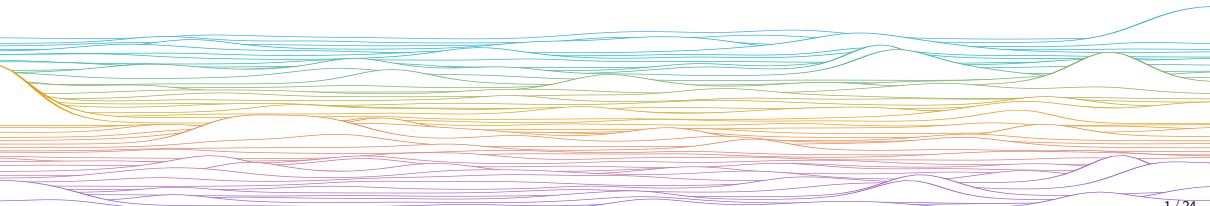
University of Bristol

Adapted in part from B Mizusaki's slides

M Rule

Learning outcomes:

- ▶ Understand how voltage triggers release of chemical neurotransmitters at the presynapse
- ▶ Work with ODE models (mass-action chemical kinetics) of synaptic release
- ▶ Understand abstract representations of synapses
- ▶ Write down a simple model for a synaptic input



Start by drawing pictures to remind ourselves of the steps in synaptic transmission.

Presynaptic

- 1 Action potential arrives
- 2 Voltage gated calcium channels open
- 3 Ca^{2+} enters the presynaptic cell
- 4 Ca^{2+} triggers vesicle fusion

Quantal release

Readily-releasable pool

- ▶ A certain amount of vesicles loaded with neurotransmitter are docked near the presynaptic membrane, and bound to associated proteins that enable rapid, calcium-triggered release.

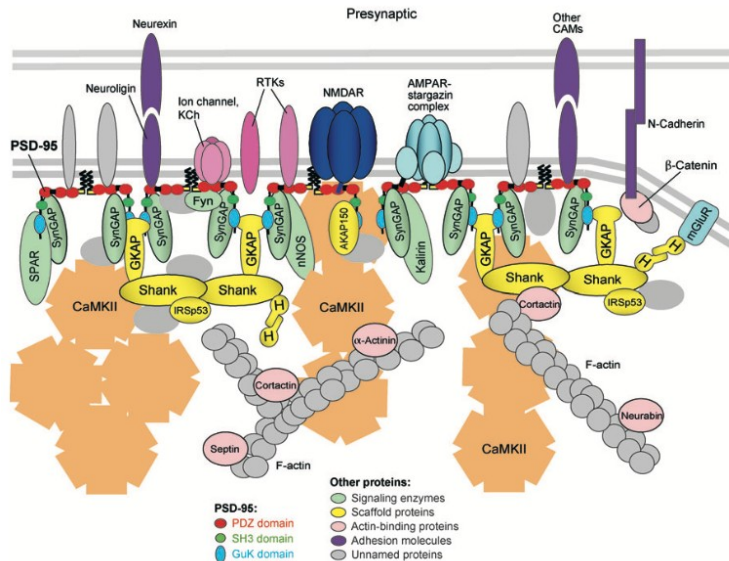
Spike arrival at axon is all-or-nothing

- ▶ Depolarisation and calcium influx triggers exocytosis of this pool

Postsynaptic

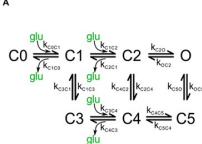
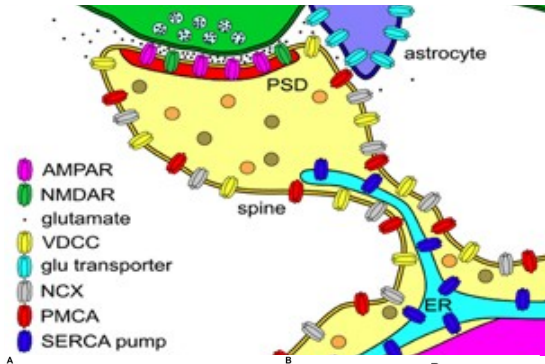
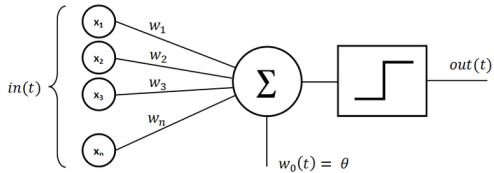
- 1 Neurotransmitter binds to receptors on postsynaptic membrane
- 2 Ionotropic receptors open, allowing ionic currents
Metabotropic receptors trigger signalling cascades that modify ionic conductances
- 3 Ionic currents change dendritic voltage

How should we model this?

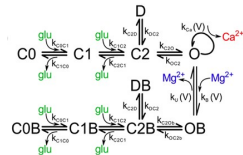


Sheng & Hoogenraad, *Annu Rev Neurosci* (2007)

Many possible levels of modelling abstraction



AMPA



NMDAR-GluN2A/2B

What goes in a model of a synapse?

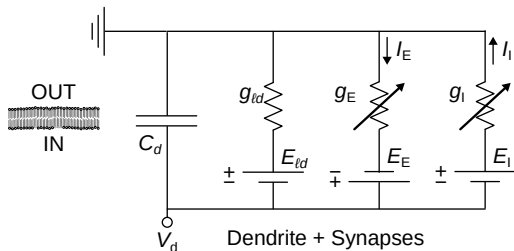
Synapses are unreliable (successful vesicle fusion, release is probabilistic)

Weighted transformation of input amplitudes, non-linear;

Excitatory or inhibitory (depending on presynaptic neuron);

A common phenomenological model

Time-dependent conductance $\bar{g}_s s(t)$ in series with a battery (voltage E_s)



$$I_s(t) = \bar{g}_s s(t) [E_s - V(t)]$$

$\bar{g}_s$: **maximal conductance** $\sim \#$ receptors in synapse $\times$ conductance when open

E_s : Reversal potential for ionotropic receptor

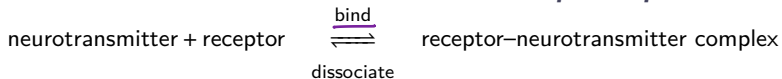
- ▶ ~ 0 for excitatory synapses (K^+ , Na^+)
- ▶ $\sim V_{rest}$ for inhibitory synapses (Cl^-)

But what is $s(t)$??

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A detailed approach: Mass-action chemical kinetics

Mass-action chemical kinetics: Ionotropic receptor



Fraction open s for fixed A ($A \gg R_{\text{tot}}$)?

$$R_{\text{tot}} := R_{\text{open}} + R_{\text{closed}}$$

total postsynaptic receptor

$$s := R_{\text{open}}/R_{\text{tot}}$$

fraction receptor bound

$$\rho_{\text{on}} = \alpha \overbrace{A R_{\text{tot}} (1-s)}^{\text{free } R}$$

Reaction rates

$$\rho_{\text{off}} = \beta R_{\text{tot}} s$$

$$\dot{s} = -\beta s + \alpha A (1-s)$$

ODE for fraction bound

$$s_{\infty}(A) = \left[1 + \frac{\beta}{\alpha A}\right]^{-1} = \sigma(\ln A - \ln \frac{\beta}{\alpha})$$

Equilibrium ($\dot{s} = 0$)



Mass-action chemical reaction kinetics: Ionotropic receptor

Model 1: Binding (ignoring clearance), solvable

$$\dot{s} = -\beta s + \alpha A(1 - s) \quad \text{ODE for fraction bound}$$

$$\dot{A} = -R_{\text{tot}} \dot{s}$$

Model 2: A added from presynapse ($u(t)$), cleared via reuptake (rate γ)

$$\dot{A} = -\gamma A + u$$

Model 1+2: Even this is too much to solve

$$\dot{s} = -\beta s + \alpha A(1 - s)$$

$$\dot{A} = -R_{\text{tot}} \dot{s} - \gamma A + u$$

release (pointing to u)
clearance (under γA)

We just need:

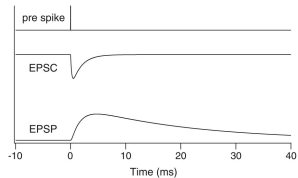
- Post-synaptic conductance goes up, then down again, whenever the presynaptic axonal terminal spikes

For presynaptic spike times $\mathcal{T}$:

$$s(t) = \sum_{t_s \in \mathcal{T}} h_s(t - t_s) \quad \leftarrow \text{linear sum}$$

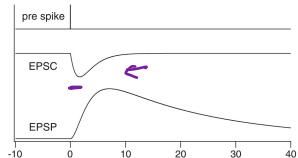
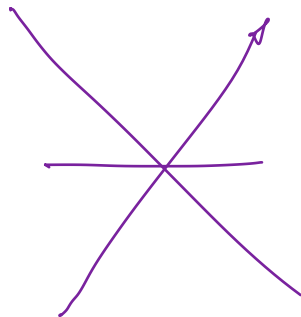
Single exponential

$$h_s(t) = e^{-t/\tau_s}$$



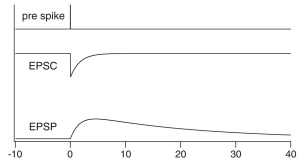
Alpha function

$$h_s(t) = te^{-t/\tau_s}$$



Difference of exponentials

$$h_s(t) = e^{-t/\tau_{\text{decay}}} - e^{-t/\tau_{\text{rise}}}$$



Roth & van Rossum

$$y(t) \quad h(t) \quad s(t) = [y * h](t)$$

↑
convolution operation



$$s(t) = \sum_{t_s} \underline{h(t - t_s)}$$



$$e^{-t/\tau_{syn}}$$

$$s(t) = \int_{-\infty}^{\infty} y(t') \underline{h(t - t')} dt'$$

$$\delta(t - t_s)$$



$$y(t)$$

$$\lim_{\sigma \rightarrow 0}$$



$$\delta(t - t_s)$$

Summary

Real synapses are complicated molecular machines

We can model them at multiple levels of granularity, as appropriate for the task at hand

A simple way of modelling a PSP on the postsynaptic membrane potential is with the addition of a single impulse to the current.

Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin–Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch–Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative

end

... Quantum Chemistry, Molecular dynamics

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